

VisionEval Workbench 2.0 macOS User Guide

Reference guide for using Workbench after installation

This guide explains how to inspect model inputs, build and copy scenarios, run VisionEval, compare datastores, manage Hypercube experiments, export results, and maintain a local Workbench workspace.

Apple Silicon macOS | Version 2.0 | September 2026

Signed and notarized Apple Silicon edition

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1 About this guide

Use this document as a reference after installing the signed and notarized VisionEval Workbench application and Docker Desktop on an Apple Silicon Mac.

VisionEval Workbench is a local desktop application for inspecting a VisionEval model, creating repeatable scenarios, running those scenarios in an isolated Docker runtime, and comparing the resulting Datastores. Version 2.0 adds a dedicated Hypercube workflow and expands project management, queue coordination, comparison, mapping, exports, storage safeguards, and background-operation reporting.

The signed disk image uses Apple's Developer ID, hardened runtime, notarization, and a stapled ticket. Drag the application to Applications and open it normally; no Gatekeeper workaround is expected. This guide begins with the first Workbench launch, follows the five primary tabs, and ends with a feature locator and troubleshooting reference.

Supported environment

- Apple Silicon Mac using the signed, notarized ARM64 Workbench application.
- Docker Desktop for Apple silicon when model runs or uncached R datastore access are required.
- The verified Workbench runtime built from official VisionEval VE-40-RC7.
- A Finder-visible workspace selected by the user. Projects and results remain separate from the application bundle.

What works without Docker

Explore, Create, workspace management, and previously cached comparison data remain available when Docker Desktop is absent or stopped. Running VisionEval requires Docker. Reading an uncached R datastore can also require the verified runtime.

2 Workbench at a glance

The primary navigation follows the normal path from understanding inputs to interpreting results.

Tab	Purpose	Typical output
1 Explore	Read input explanations and inspect declared model dependencies.	Input understanding and dependency diagrams
2 Create	Develop regions, create projects, edit scenarios, and review saved changes.	Validated project and scenario definitions
3 Run	Queue model work and inspect status, logs, history, and results.	Verified VisionEval Datastores
4 Compare	Inspect observed differences, maps, charts, and exports.	Tables, maps, charts, and analysis files
5 Hypercube	Build, run, and analyze bounded parameter matrices.	Case batches and response analysis

What Workbench manages

Workbench manages the workspace you choose, its projects, Workbench run containers, registered results, and temporary comparison data. It does not edit the original package source or stop unrelated Docker containers.

What remains authoritative

The Datastore created by a successful run is the official result. Workbench also creates temporary indexes and summaries to make Compare and Hypercube analysis faster. You can clear and rebuild this temporary data without deleting the Datastore.

3 First start after installation

Complete these checks before creating production projects.

1. Choose or open a workspace. Workbench stores assets, projects, jobs, results, caches, documentation, and notes there.
2. Confirm the runtime status at the top of the window or in Settings. Ready means Docker is running and the approved runtime has passed verification.
3. Open Settings and add a model package. Workbench validates the package and displays its type, version, contents, compatibility, provenance, intended use, execution support, warnings, and checksum status before installation.
4. Open Explore and select the installed package. Confirm that its input explanations and dependency metadata load.
5. Create a small standard project and review it before attempting a large batch or Hypercube.

Package installation safety

Selecting a ZIP or extracted folder does not immediately change the workspace. Workbench validates the source, shows a preview, and installs only after confirmation. It validates the source again immediately before copying so a changed package cannot bypass the preview.

The Virginia package is source data for building supported MPO or custom regional packages. It is not intended to run Virginia as one statewide model.

4 Core concepts

These terms describe the objects that appear throughout the interface.

Term	Meaning
Workspace	The user-selected folder containing Workbench-managed projects, assets, results, logs, caches, and documentation.
Package	A validated ZIP or folder that can provide a runnable model, InputLibrary, explanations, regions, map context, or package documentation.
Project	A baseline plus scenarios or a Hypercube matrix, tied to a specific installed model package.

Term	Meaning
Scenario	A named set of deliberate input changes layered over a project's untouched baseline.
Run	One prepared VisionEval execution with recorded project, scenario, runtime, status, timestamps, and logs.
Result	A registered successful Datastore with provenance and compatibility information.
Azone	A VisionEval area geography. In Virginia packages, Azones map to county-equivalent localities.
Bzone	A smaller VisionEval geography used for regional definitions and detailed mapping.
Marea	A logical metropolitan-area geography. Supported Virginia regional packages can group detailed zones into named Marea regions.
Runtime profile	The verified Docker adapter, architecture, immutable image digest, VisionEval version, and verification time.

Scenario calculations

Ordinary batch operations apply on top of the scenario's current values. Select Start this batch from the untouched baseline when the selected files, columns, year, and locations should be recalculated from the project's original baseline instead. The interface describes the active mode directly before the operation is saved.

5 Explore

Explore explains available inputs and declared module relationships before you decide what to change.

Input File Library

Choose an installed model package, then search by file, field, or plain-language topic. Workbench combines package-provided explanations with built-in VisionEval metadata. Select a file to see its purpose, fields, units, allowed values, years, and related modules when that metadata is available.

Dependencies

The dependency canvas displays relationships declared by the selected model. Use search and focus controls to isolate a module or input path. Export a focused diagram as SVG or PDF when it supports review or documentation.

A dependency path shows that a model component can use an input. It does not prove that a particular scenario change caused an observed numerical difference. Compare completed Datastores to inspect actual results.

6 Create and review

Create holds standard scenario development. Hypercube projects are created only in the dedicated Hypercube tab.

Develop

Develop builds a regional package from an eligible installed model or regional-data package. Select a supported predefined region or choose planner-defined geographies, inspect the preview, and build only after validation succeeds. The first map view may retrieve official geometry and cache it; region preview and package building use the packaged crosswalk and can remain available offline.

Setup

Create a standard project by naming it, choosing a model package, and choosing whether to create a fresh untouched baseline or reuse a verified compatible completed baseline. Scenario names are unique across the project without regard to capitalization. A cross-project copy keeps its original name when available; collisions become Name (copy), Name (copy 2), and so forth. Copied results and provenance use the resolved destination name.

Editor

Use single-file mode for direct table edits and Batch Change for one operation across several files, columns, years, or locations. The sidebar records saved files beneath each scenario. Creation, duplication, renaming, Hypercube generation, and cross-project copying all enforce case-insensitive project-wide name uniqueness.

- Single-file mode shows the source schema, editable values, filters, and overlap confirmation when a new edit replaces an existing change.
- Batch Change lets you select eligible files and columns, choose location scope and target year, enter an operation, and save the changes together.
- The starting-values bar makes compounding behavior explicit. Current scenario values add to saved edits; untouched baseline values replace earlier changes only within the selected scope.
- Draft Batch Change values remain available while you move between Create views during the current application session.

Review

Review is the release gate for a scenario. It lists saved file changes, replacement or compounding behavior, target geography and year, validation results, and provenance. Continue to Run opens the Run tab with the reviewed project and scenarios selected; it never starts a model automatically.

A scenario with an unfinished run cannot be removed. Removing a standard scenario previews the files, terminal runs, logs, completed results, and storage that will be deleted. Generated Hypercube cases cannot be removed individually because they belong to one matrix definition.

7 Run

Run prepares model copies, queues VisionEval work, shows live output, and registers verified results.

Before running

- Docker Desktop is running and the Workbench runtime reports Ready.
- The project uses an installed model package that matches its saved identity.

- The selected scenario has no blocking Review errors.
- Available disk space and Docker memory are appropriate for the selected work.

Queue and execution

Choose a standard project and explicitly select the baseline or scenarios to run. Normal and Hypercube work share one workspace FIFO queue. The oldest unfinished batch owns the available runtime slots; its jobs may run concurrently, while later batches stay grouped and begin automatically in order. Parallel jobs share Docker Desktop's total memory allocation. Workbench's optional per-run memory limit is a ceiling, not a reservation.

History and live output

Run History has its own scrollbar so a long history does not extend the page. Standard jobs remain individual cards. Each Hypercube project appears as one compact card showing active slots, current case range, progress, waiting work, and approximate time remaining; detailed case logs stay in Hypercube Run. Select a standard job to inspect its R console output and status details.

Stopping retrying and cleanup

Stop Run requests cancellation of one selected standard job; Stop All Runs applies to Workbench-owned active and queued jobs. Stop This Hypercube removes waiting work and stops only owned active containers for the selected Hypercube, leaving unrelated batches untouched. Cleanup verifies workspace, job, and execution-attempt ownership before removing a container or temporary model.

Exit code 137 can indicate memory pressure. Workbench reports confirmed exhaustion only when Docker marks the container OOMKilled. Reduce concurrency, increase Docker Desktop's allocation, or review an overly restrictive per-run limit before retrying.

History retention

Clear history previews the number of terminal jobs, logs, and removable bytes. It preserves projects, scenarios, Datastores, registered results, and comparison data. Terminal history and logs older than 30 days are removed automatically under the same active-job protections.

8 Compare

Compare lets you examine completed results side by side, view maps and charts, and save tables or graphics.

Selections in each view

Compare, Map Visualization, and Percent-Change Chart each remember their own selected results. The first view you open can use the most recent Reference and Comparison; later choices remain local to that view. Regular selectors show standard and imported results. Hypercube analysis remains in the dedicated Hypercube tab, with Open Map available for a selected case when the output supports spatial display.

Compare values

Choose a Reference result and, if needed, a Comparison result. Then choose the table, variable, year, number of rows, and location level. With only a Reference, you can inspect and export its values. Choose Update Comparison to calculate differences between two results.

Find All Changes

Choose Find All Changes to scan every comparable output across the selected Reference and Comparison. It is independent of the detailed Table and Variable controls, reports progress, supports cancellation, and can reuse safely fingerprinted cached work when the selected roles, data, year, geography, and filters are unchanged.

Map Visualization

Generate Map shades package-defined geography using the palette saved in Settings. Change measures use a symmetric zero-centered range based on the largest visible absolute change. Reference and comparison measures use the visible minimum and maximum. A perceptual intensity ramp keeps smaller nonzero values visible while exact zero remains neutral.

- Two-dimensional and three-dimensional maps share the same values, palette, scale limits, and selection behavior.
- Fit Project adds an inset on all sides and recalculates after the canvas size changes.
- The Virginia outline remains visible beneath the result layer; optional MPO and technical context remains subdued.
- PlanRVA Marea results display exactly two clean regions, Richmond UZA and Non-UZA, with technical membership retained for inspection and export.
- PDF, PNG, and SVG exports preserve the visual map. CSV and Excel exports preserve values and provenance.

Percent Change Chart

Choose numeric outputs and generate a diverging horizontal bar chart for one year and location scope. Decreases extend left of zero and increases extend right. After generation, sort and display controls transform the cached result without recalculating the Datastores.

Exports

Current View exports all rows matching the active variable and filters, not only the visible page. Full Variable Data exports complete comparison-compatible outputs in original order. CSV creates a ZIP with one file per output, while Excel creates typed worksheets with statistics and provenance. All user-initiated file exports, including diagrams and diagnostics, pass through one application-wide FIFO queue so save dialogs occur predictably.

9 Hypercube

Hypercube runs many complete VisionEval cases and should be used on a dedicated computer or a suitably sized cloud virtual machine.

Open primary tab 5 or press Command 5. Workbench presents a resource disclosure once per application launch before any Hypercube subtab is shown. Its fixed 81-case Apple Silicon planning example renders immediately; Build and Review use measured history when compatible successful runs are available. Review run count, execution waves, estimated time, memory range, and retained disk before generating cases.

Build

Create or select a Hypercube project and define one or two numeric axes. A new or replacement axis requires an interval of at least 2 and may generate no more than 20 values. A matrix may contain no more than 400 cases. Existing older matrices remain runnable and analyzable even if they exceed these limits, but they cannot be replaced by another out-of-policy definition.

Draft definitions autosave. Preview Matrix shows the exact generated values, case count, shared scope, waves, runtime estimate, memory range, and retained disk. Generate Scenarios creates or replaces the cases only after the preview is valid.

Hypercube Build separates project creation from matrix definition

Review

Review shows the saved definition, generated cases, file and row scope, provenance, and resource estimate. Use it to verify that every case changes only the intended inputs before model work is queued.

Run

Choose the project rather than individual cases. Run Missing queues only a missing baseline and cases; Retry Failed queues only failures. The batch card reports counts, progress, execution waves, concurrency, a measured or planning-based ETA, and elapsed time. Elapsed time freezes when each batch reaches a terminal state and excludes idle gaps between retry batches. Stop This Hypercube affects only the selected project. Hypercube completion produces one project-level notification rather than one alert per case.

Analyze

Analyze provides a cached Response Matrix, Scenario ranking, optional one- or two-cell details, Open Map for supported spatial outputs, and Find All Changes. Response Curves and general Compare handoffs are not part of Hypercube Analysis. Choose Sum, Mean, Median, Minimum, or Maximum explicitly for the Response Matrix; there is no automatic aggregation. Selecting a matrix cell uses the completed result immediately and does not recalculate the matrix.

- Response Matrix request controls lock only while a new fingerprinted matrix is being calculated. The previous completed matrix remains visible during an update, and returning to an identical request reuses the disposable cache.
- The activity bar shows the current phase, case or table progress, elapsed time, heartbeat, bytes read, summaries written, and cache growth, with cancellation when supported.

- Find All Changes has its own Year and Aggregation controls and is independent of the Response Matrix table, variable, metric, geography, locations, axes, and aggregation. Median is its default; Mean, Sum, Minimum, and Maximum remain available.
- Find All Changes scans every eligible output and verified completed case, then offers Outputs and Scenarios views. Output headings include keyboard-accessible explanations. Selecting an output opens a side panel ranking cases by largest change, increase, decrease, proximity to baseline, or case number. Scenario summaries use the same discovery result and do not rescan Datastores.
- Identical Find All Changes requests reconnect to active work or reuse completed cache entries. Completion and failure can notify you while another tab is visible. Rankings describe sensitivity and deviation from baseline, not whether an outcome is desirable.

Hypercube storage

Hypercube runs retain the authoritative Datastore and do not create persistent full CSV trees. Response matrices, Find All Changes summaries, and extracted columns are disposable analysis cache entries shared with Compare under the 5 GB cache limit. Clearing the cache never deletes a project or Datastore.

Export

Use the Export subtab to select the common baseline or up to three verified completed cases per batch. Each item becomes a separate ZIP containing resolved input CSVs, output CSVs generated temporarily from its Datastore, and a manifest; the Datastore itself is excluded. The activity panel reports package position, extraction, compression, save-dialog state, cleanup, size, heartbeat, and queued count. Cancelling or failing one package does not block later FIFO exports, and temporary CSVs are removed after every outcome.

10 Virginia regional data

The Virginia source-data package helps create supported MPO or custom regional model packages and provides statewide map context.

What the Virginia package provides

The package includes the statewide Virginia InputLibrary, definitions for Virginia MPO study areas, and the information Workbench needs to connect those study areas with VisionEval zones. Use Create and then Develop to preview a supported MPO or choose a custom group of Virginia localities before building a regional package.

Runnable regional packages

Packages such as PlanRVA and WPPDC contain the model inputs for their own regions and are designed to run as regional models. Available packages may change over time, so this guide does not list each package in detail. Use Explore and the package preview in Settings to see what an installed package contains and whether it supports model runs.

Statewide limitation

The statewide package provides source data and map context; it is not a supported statewide runnable model. Build and run an MPO or custom regional package instead. If someone tries to run the statewide inputs directly, a model-data error can occur. Reinstalling Docker or repeating the same run will not fix that condition.

11 Settings workspaces and storage

Settings controls application preferences and managed resources without editing project data directly.

Workspace and assets

Workspace shows the active location and provides supported ways to open or move between workspaces. Assets lists installed packages and begins the validated package-preview flow. Use Workbench rather than Finder to remove managed assets or project data.

Appearance numbers and notifications

Appearance controls theme and visualization palettes. Number settings control display precision without changing stored values. Notifications can report completed or failed long-running work, including Compare and Hypercube operations that finish while another tab is visible. Hypercube case runs stay quiet until the project batch is complete.

Runtime and resources

Runtime begins with Ready, Needs attention, or Not configured. The summary shows the VisionEval version, computer architecture, last verification, and main actions. More technical image and verification details are available in expandable sections when needed.

Measured MPO or regional guidance is approximately 2.5 to 3.5 GB of Docker memory per active run. Actual use depends on population, model stages, years, and outputs. These values guide planning but do not guarantee a run will fit.

Updates

Update checks are off by default. You can run them manually or allow a weekly check. Workbench never installs an application or runtime update without confirmation. Before a runtime update is used, Workbench verifies it and keeps the previous verified runtime available for rollback.

Storage

Storage separates authoritative Datastores, optional Standard-project full CSV exports, disposable comparison and Hypercube analysis caches, logs, and other managed data. Standard runs follow the Retain full VisionEval CSV exports preference recorded when the job is queued. Hypercube runs always retain Datastores without persistent full CSV trees.

Reset

Reset settings restores preferences such as appearance, numbers, notifications, export retention, resource limits, and Docker-start behavior. It preserves workspaces, packages, projects, results, runtime profiles, and runtime images.

12 Accessibility and keyboard use

Workbench supports keyboard navigation, focus indication, descriptive control names, and reduced reliance on color alone.

Shortcut	Action
Command 1	Open Explore
Command 2	Open Create

Shortcut	Action
Command 3	Open Run
Command 4	Open Compare
Command 5	Open Hypercube
Command comma	Open Settings
Tab and Shift Tab	Move through interactive controls
Space or Return	Activate the focused button, checkbox, or selection
Escape	Close a menu or dismissible dialog when supported

Disabled analysis controls sit inside a focusable lock wrapper that explains why selections cannot change. Map and chart values use labels, signs, outlines, or patterns in addition to color. Long image identities and digests remain available to assistive technology even when shortened visually.

13 Provenance units and interpretation

Workbench preserves enough context to reproduce and audit a scenario without turning a dependency or comparison into a causal claim.

Provenance

Each project records which model package and InputLibrary it uses. Each run adds the project, scenario, Workbench version, runtime version, date, status, and result check. Exports record the selected results, year, table, variable, units, locations, and creation time. Workspace paths are not added to exports.

Units and rounding

Input explanations and output metadata identify units when the selected package or VisionEval metadata provides them. Display rounding never changes stored values. Comparisons calculate from source values and then format the result for the interface or export.

Row identity

Household, Vehicle, and Worker row identifiers can change between runs. For that reason, Find All Changes compares aggregate distributions for those tables instead of assuming that the same row represents the same person, household, or vehicle in every run. Detailed rows remain available when an output supports safe matching.

Interpreting maps

A map summarizes values at the selected geography and aggregation. Technical identifiers remain in inspectors and exports. Missing or undefined percent changes use an unavailable pattern rather than an invented zero. A visible state or MPO outline provides context and does not mean that every outlined area contributed to the result.

14 Safe shutdown recovery and maintenance

Workbench coordinates background operations before closing and preserves restart-safe records when cleanup cannot finish.

Quit protection

If runs, Hypercube generation, matrix calculations, Find All Changes, indexing, comparisons, exports, project copies, regional builds, or runtime installation are active, the quit dialog lists the operation types and counts. Keep Working leaves them running. Stop Work and Quit requests cancellation, stops only Workbench-owned containers, flushes autosaves, and closes only after safe cleanup succeeds.

Recovery

Queued and active jobs keep restart-safe manifests. On launch, Workbench reconciles those records with owned containers and partial outputs. Cleanup failures remain actionable rather than being hidden. Comparison and Hypercube summary caches can be discarded and rebuilt from authoritative Datastores.

Routine maintenance

- Keep adequate free disk space for model copies and Datastores.
- Review Docker Desktop's total memory allocation before increasing run concurrency.
- Clear terminal run history when it is no longer useful; results remain registered.
- Clear comparison caches when storage is needed or a disposable index appears stale.
- Use Settings Diagnostics to collect a support bundle for a repeatable failure.

15 Troubleshooting

Start with the visible status and preserve the exact error before changing the workspace or runtime.

Problem	What to check
Workspace is not found	Open the original workspace folder. Do not create a replacement inside an existing managed workspace.
Docker is stopped	Start Docker Desktop from Workbench or macOS, wait for the engine, then verify the runtime again.
Runtime verification fails	Open Settings and then Runtime to review the failed checks. Confirm that Docker is running, the Apple Silicon runtime is selected, and the expected version is installed. Then choose Verify runtime again.
Project is absent from Run	Confirm it is a standard project, its model package is installed, and Review has no blocking errors.
Scenario change is missing	Confirm the saved file, target year, location scope, and starting-value mode in Create Review.

Problem	What to check
Find All Changes is slow	A first uncached scan reads every eligible output and completed case. Keep the activity strip open; an 81-case regional scan is often around 10 minutes on Apple Silicon, while larger data can take longer. Repeating the same year and aggregation can reuse the cache.
Stop leaves cleanup failed	Start Docker if required and select Retry Cleanup. A result is not registered until cleanup is complete.
Exit code 137	Check the job's failure type. Workbench calls exit 137 memory-related only when Docker reports OOMKilled=true; otherwise it may reflect cancellation or an external stop. Adjust concurrency or Docker memory only when the evidence indicates pressure.
Virginia statewide run fails	Use an MPO or custom regional package. The Version 2.0 statewide input set is not supported for a complete run.

Support information

For a run failure, preserve the failed job log and export a diagnostics bundle. Record the Workbench version, macOS version, installed package versions, runtime status, and the exact operation that failed. Do not include confidential input data in a public issue.

Issue tracker github.com/nikolasleeb/VisionEval-Workbench/issues

16 Feature locator and glossary

Use this section to find the principal command or setting associated with a task.

Task	Location
Understand an input field	Explore Input File Library
Inspect declared dependencies	Explore Dependencies
Build an MPO or custom region	Create Develop
Create a standard project	Create Setup
Edit one input table	Create Editor Single file
Apply one operation to several inputs	Create Editor Batch Change

Task	Location
Verify saved scenario changes	Create Review
Run a standard baseline or scenario	Run Select runs
Inspect R output	Run Live output
Clear terminal run history	Run History Clear history
Compare one output	Compare Compare
Find all changes between two results	Compare Find All Changes
Find changes across a Hypercube	Hypercube Analyze Find All Changes
Generate a map	Compare Map Visualization
Generate a multi-output chart	Compare Percent-Change Chart
Create a parameter matrix	Hypercube Build
Run missing Hypercube cases	Hypercube Run Run Missing
Analyze a response matrix or rank scenarios	Hypercube Analyze
Install or inspect a package	Settings Assets
Verify or roll back the runtime	Settings Runtime
Review memory and concurrency	Settings Resources
Clear disposable caches	Settings Storage
Export support information	Settings Diagnostics

17 Current limitations

These boundaries prevent the interface from implying support that the underlying model or release has not demonstrated.

- Version 2.0 documentation and this signed, notarized release cover Apple Silicon macOS. Intel macOS and Windows remain separate implementations.
- The macOS application uses Docker Desktop and the approved ARM64 runtime. Native R execution is not supported by this build.

- A complete statewide Virginia execution is not supported. Use MPO or custom regional packages for runnable scenario work.
- Hypercube resource estimates are planning guidance. Actual duration, memory, and disk use depend on the selected model and computer.
- Dependency diagrams describe declared relationships. Only completed Datastore comparisons show observed numerical differences.
- Disposable caches improve performance but are not substitutes for Datastores or run provenance.
- The application is local-first. Update checks and official map downloads use the network only for their stated actions.

VisionEval documentation visioneval.github.io/docs

Workbench repository github.com/nikolasleeb/VisionEval-Workbench